

Chapter 4: Integers and Rationals

Analysis I, by Terrence Tao

A: The Integers

Problem 1 - Equality on the integers follow the three axioms

Verify that the definition of equality on the integers is reflexive, symmetric, and transitive.

Proof. First, we show that the equality is reflexive. Consider the integer $a \text{---} b$. Notice that $a + b = a + b$, so $a \text{---} b = a \text{---} b$.

Second, we prove that the equality is symmetric. Suppose $a \text{---} b = a' \text{---} b'$. Then, $a + b' = a' + b$. Because the equality on the natural numbers is transitive, we have that $a' + b = a + b'$. By definition of the equality on the integers, we have that $a' \text{---} b' = a \text{---} b$.

Next, we show that the equality on the integers is symmetric. Suppose $a \text{---} b = c \text{---} d$ and $c \text{---} d = e \text{---} f$. Then, $a + d = c + b$ and $c + f = e + d$. We add the second equation into the first to get:

$$a + d + c + f = c + b + e + d.$$

By the cancellation law of natural numbers for addition, we can cancel out c and d to get:

$$a + f = b + e.$$

By the definition of equality on the integers, $a \text{---} b = e \text{---} f$. □

Problem 2 - Negation on the Integers is well-defined

Show the definition of negation on the integers is well-defined in the sense that if $(a \text{---} b) = (a' \text{---} b')$, then $-(a \text{---} b) = -(a' \text{---} b')$. Hence, equal integers have equal negations.

Proof. Suppose $(a \text{---} b) = (a' \text{---} b')$, so $a + b' = a' + b$. Because addition of the natural numbers is commutative, we have that $b' + a = b + a'$. Then, by the symmetry of the equality of the natural numbers, we have $b + a' = b' + a$. Then, by the definition of the equality on the integers, we have $b \text{---} a = b' \text{---} a'$. Because $b \text{---} a = -(a \text{---} b)$ and $b' \text{---} a' = -(a' \text{---} b')$, we get:

$$-(a \text{---} b) = -(a' \text{---} b').$$

□

Problem 3 - Multiplying by -1 is the same as negation

Show that $(-1) \times a = -a$ for every integer a .

Proof. Recall that $-1 = 0 \text{---} 1$ and $a = (a \text{---} 0)$. Notice:

$$\begin{aligned} (-1) \times a &= (0 \text{---} 1) \times (a \text{---} 0) \\ &= (0 \times a + 1 \times 0) \text{---} (0 \times 0 + 1 \times a) \\ &= (0 + 0) \text{---} (0 + a) \\ &= 0 \text{---} a \\ &= -a \end{aligned}$$

□

Problem 4 - Properties of the Integers

Prove the remaining identities in Proposition 4.1.6. For all of the laws of algebra for the integers, let $x = x_1 - x_2, y = y_1 - y_2, z = z_1 - z_2$ be integers, where $x_1, x_2, y_1, y_2, z_1, z_2 \in \mathbb{N}$.

Proposition 4.1. $x + y = y + x$

Proof. Notice:

$$\begin{aligned} x + y &= (x_1 - x_2) + (y_1 - y_2) \\ &= (x_1 + y_1) - (x_2 + y_2) \\ &= (y_1 + x_1) - (y_2 + x_2) \\ &= y + x. \end{aligned}$$

□

Proposition 4.2. $(x + y) + z = x + (y + z)$

Proof. Notice:

$$\begin{aligned} (x + y) + z &= ((x_1 - x_2) + (y_1 - y_2)) \\ &\quad + (z_1 - z_2) \\ &= ((x_1 + y_1) - (x_2 + y_2)) \\ &\quad + (z_1 - z_2) \\ &= (x_1 + y_1 + z_1) - (x_2 + y_2 + z_2) \\ &= (x_1 + (y_1 + z_1)) \\ &\quad - (x_2 + (y_2 + z_2)) \\ &= (x_1 - x_2) \\ &\quad + ((y_1 + z_1) - (y_2 + z_2)) \\ &= x + (y + z). \end{aligned}$$

□

Proposition 4.3. $x + 0 = 0 + x = x$

Proof. Recall that $x = (x - 0)$ and $0 = (0 - 0)$. Notice:

$$\begin{aligned} x + 0 &= (x - 0) + (0 - 0) \\ &= (x + 0) - (0 + 0) \\ &= x - 0 \\ &= x. \end{aligned}$$

Similarly, because addition is commutative, we have that $0 + x = x$. □

Proposition 4.4. $x + (-x) = (-x) + x = 0$

Proof. Notice:

$$\begin{aligned} x + (-x) &= (x_1 - x_2) \\ &\quad + (x_2 - x_1) \\ &= (x_1 + x_2) - (x_2 + x_1) \\ &= (x_1 + x_2) - (x_1 + x_2) \\ &= 0. \end{aligned}$$

Because addition is commutative, we have that $(-x) + x = 0$. □

Proposition 4.5. $xy = yx$

Proof. Notice:

$$\begin{aligned} xy &= (x_1 - x_2) \times (y_1 - y_2) \\ &= (x_1y_1 + x_2y_2) - (x_1y_2 + x_2y_1) \\ &= (y_1x_1 + y_2x_2) - (y_2x_1 + y_1x_2) \\ &= (y_1x_1 + y_2x_2) - (y_1x_2 + y_2x_1) \\ &= yx. \end{aligned}$$

□

Proposition 4.6. $(xy)z = x(yz)$.

Proof. This has already been done in the book. □

Proposition 4.7. $x1 = 1x = x$

Proof. Recall that $1 = (1 - 0)$. Notice:

$$\begin{aligned} 1x &= (1 - 0) \times (x_1 - x_2) \\ &= (1x_1 + 0x_2) - (1x_2 + 0x_1) \\ &= x_1 - x_2 \\ &= x. \end{aligned}$$

Because multiplication is commutative, $x1 = x$. □

Proposition 4.8. $x(y + z) = xy + xz$

Proof. Notice:

$$\begin{aligned} x(y + z) &= (x_1 - x_2)((y_1 - y_2) + (z_1 - z_2)) \\ &= (x_1 - x_2)((y_1 + z_1) - (y_2 + z_2)) \\ &= (x_1(y_1 + z_1) + x_2(y_2 + z_2)) \\ &\quad - (x_1(y_2 + z_2) + x_2(y_1 + z_1)) \\ &= (x_1y_1 + x_1z_1 + x_2y_2 + x_2z_2) \\ &\quad - (x_1y_2 + x_1z_2 + x_2y_1 + x_2z_1) \\ &= ((x_1y_1 + x_2y_2) + (x_1z_1 + x_2z_2)) \\ &\quad - ((x_1y_2 + x_2y_1) + (x_1z_2 + x_2z_1)) \\ &= ((x_1y_1 + x_2y_2) - (x_1y_2 + x_2y_1)) \\ &\quad + ((x_1z_1 + x_2z_2) - (x_1z_2 + x_2z_1)) \\ &= xy + xz \end{aligned}$$

□

Proposition 4.9. $(y + z)x = yx + zx$

Proof. Notice:

$$\begin{aligned} (y + z)x &= x(y + z) \\ &= xy + xz \\ &= yx + zx. \end{aligned}$$

□

Problem 5 - Integers have no zero divisors

Let $a, b \in \mathbb{Z}$. Suppose $ab = 0$. Then, $a = 0$ or $b = 0$.

Proof. Suppose that $ab = 0$. We will use the trichotomy of the integers on the variable a and the cancellation law of the naturals. We will prove the proposition for all three cases of $a = 0$, a is a positive integer, and a is a negative integer. In all these cases, b is held fixed with $b = b_1 - b_2$, where $b_1, b_2 \in \mathbb{N}$.

1. Say $a = 0$. Then, the proposition is trivially true.
2. Say a is a positive number, so a takes on the form $a = k - 0$ for some positive natural number k . Then, $(k - 0) \times (b_1 - b_2) = (kb_1 + 0b_2) - (kb_2 + 0b_1) = kb_1 - kb_2$. Because $ab = 0$, we have that $(kb_1 - kb_2) = (0 - 0)$, which means that $kb_1 = kb_2$. By the cancellation law of the naturals, we have that $b_1 = b_2$, so $b = 0$.
3. Say a is a negative number, so a takes on the form $a = 0 - k$. Then, $(0 - k) \times (b_1 - b_2) = (0b_1 + kb_2) - (0b_2 + kb_1) = kb_2 - kb_1$. Because $ab = 0$, we have $(kb_2 - kb_1) = 0$, so $kb_2 = kb_1$. By the cancellation law, we have that $b_2 = b_1$, so $b = 0$.

□

Problem 6 - Cancellation law for integers

Let $a, b, c \in \mathbb{Z}$ with $c \neq 0$. If $ac = bc$, then $a = b$.

There are two ways to prove this. The first proof uses Prop 4.1.8 where it states that integers have no zero divisors.

Proof. Suppose $ac = bc$. Notice:

$$\begin{array}{ll}
 ac - bc = 0 & \text{by subtracting } bc \text{ to both sides} \\
 c(a - b) = 0 & \text{by distributing } c \\
 a - b = 0 & \text{by Prop. 4.1.8 since } c \neq 0 \\
 a = b. & \text{by adding } b \text{ to both sides}
 \end{array}$$

□

The second proof uses the trichotomy of the integers on the variable c , and the cancellation law of multiplication for the natural numbers to eventually remove k in the equations.

Proof. We fix $a = a_1 - a_2$ and $b = b_1 - b_2$. Since $c \neq 0$, we only consider two cases.

1. Say c is a positive number, so there exists a positive natural number k such that $c = k - 0$. Notice:

$$\begin{aligned}
 ac = bc &\implies (a_1 - a_2)(k - 0) = (b_1 - b_2)(k - 0) \\
 &\implies ka_1 - ka_2 = kb_1 - kb_2 \\
 &\implies ka_1 + kb_2 = kb_1 + ka_2 \\
 &\implies k(a_1 + b_2) = k(b_1 + a_2) \\
 &\implies a_1 + b_2 = b_1 + a_2 \\
 &\implies a_1 - a_2 = b_1 - b_2 \\
 &\implies a = b.
 \end{aligned}$$

2. Say c is a negative number, so c is the negation $-k$ of a positive natural number k . In other words, $c = 0 - k$. We get a very similar proof to the second case of c being a positive number. Notice:

$$\begin{aligned}
 ac = bc &\implies (a_1 - a_2)(0 - k) = (b_1 - b_2)(0 - k) \\
 &\implies ka_2 - ka_1 = kb_2 - kb_1 \\
 &\implies k(a_2 + b_1) = k(b_2 + a_1) \\
 &\implies a_2 + b_1 = b_2 + a_1 \\
 &\implies a_1 - a_2 = b_1 - b_2 \\
 &\implies a = b.
 \end{aligned}$$

□

Lemma 1. Let $a, b, k \in \mathbb{Z}$. If $a = b + k$, THEN $a = b$ if and only if $k = 0$.

Proof. Suppose $a = b + k$. Let us prove the forward direction. Assume $a = b$. Then $a = b = b + k$. We subtract b to all sides to get $0 = k$. To prove the converse, assume that $k = 0$. Trivially, $a = b + k = b + 0 = b$. □

Problem 7 - Properties of order for the integers

Prove the properties of order for the integers. For all the propositions, let $a, b, c \in \mathbb{Z}$.

Proposition 7.1. $a > b$ if and only if $a - b$ is a positive number.

Proof. Suppose $a > b$. By definition of order, there exists some natural number k such that $a = b + k$ and $a \neq b$. Hence, by Lemma 1, $k \neq 0$, and k therefore is a positive number. In the other direction, suppose $a - b$ is a positive number. Define $k := a - b \neq 0$, so $a = b + k$, and hence $a > b$. By Lemma 1, since $k \neq 0$, we have that $a \neq b$. Therefore, $a > b$. □

Proposition 7.2. Addition preserves order. In other words, if $a > b$, then $a + c > b + c$.

Proof. Suppose $a > b$. Then, $a = b + k$ for some positive number k . Adding c to both sides of the equation gives $(a + c) = (b + c) + k$. Hence, $a + c > b + c$. □

Proposition 7.3. Positive multiplication preserves order. If $a > b$ and c is positive, then $ac > bc$.

Proof. Suppose $a > b$, so $a = b + k$ for some positive number k . Notice $ac = (b + k)c = bc + kc$. Since $c \neq 0$ and $k \neq 0$, we have $kc \neq 0$. Therefore, $ac > bc$. □

Proposition 7.4. Negation reverses order. If $a > b$, then $-a < -b$ (i.e., $-b > -a$).

Proof. Suppose $a > b$, so $a = b + k$ for some positive number k . We take the negation to get $-a = -b - k$. Notice that $-b = -a + k$, so $-b > -a$. □

Proposition 7.5. Order is transitive. If $a > b$ and $b > c$, then $a > c$.

Proof. Suppose $a > b$ and $b > c$, so $a = b + k$ and $b = c + l$ for positive numbers k and l . Notice, $a = b + k = c + l + k = c + (l + k)$. Since $l + k$ is also a positive number, we have $a > c$. □

Proposition 7.6. Order is trichotomous. Exactly one of the statements hold $a > b$, $a < b$, or $a = b$ is true.

Proof. Consider any two integers a and b . By the trichotomy of the integers, we have exactly one of these to be true: $a - b > 0$, $a - b < 0$, or $a - b = 0$. We consider each case and how it relates to the proposition.

1. Say $a - b > 0$, so $0 + k = (a - b)$ for some positive number k . By rearrangement, $b + k = a$, so $a > b$.
2. Say $a - b < 0$, so $(a - b) + l = 0$ for some positive number l . By rearrangement, $a + l = b$, so $a < b$.
3. Say $a - b = 0$. Then, $a = b$.

□

Problem 8 - Principle of Induction is only for the natural numbers

Show that the principle of induction does not apply directly to the integers. More precisely, given an example of a property $P(n)$ pertaining to an integer n such that $P(0)$ is true, and that $P(n)$ implies $P(n++)$ for all integers n , but that $P(n)$ is not true for all integers n . Thus, induction is not as useful a tool for dealing with the integers as it is with the natural numbers.

Proof. Consider the property $P(n) : n \geq 0$. Trivially, $P(0)$ is true. To show that the inductive hypothesis holds, suppose that $P(n)$ is true for any integer n , so $n \geq 0$. Clearly, $n++ \geq 0$, so $P(n++)$ holds true. However, $P(-1)$ is false.

The more interesting question though posed by Issa Rice is “What must the solution space look like?” More precisely, we can ask “Is there some simple characterization of every property P that works as a solution to this exercise?” The answer is that for any property P , its solution space will look the same as the set of all integers greater than some non-positive integer m . In other words, $\{n \in \mathbb{Z} : P(n)\} = \{n \in \mathbb{Z} : n \geq m\}$ for some $m \leq 0$. □

B: The Rationals

Problem 1 - Properties of absolute value and distance

Verify that the definition of equality on the rationals is reflexive, symmetric, and transitive.

Proof.

□

Problem 2 - Properties of ε -closeness

Let $x, y, z \in \mathbb{Q}$. Then, the following properties hold:

Proposition 2.1. $x + y = y + x$

Proof. Chuchu

□

Problem 3 -

Let $x \in \mathbb{Q}$. We show the trichotomy of $\mathbb{Q}$, where exactly one of the following statements hold: $x = 0$, x is a positive rational number, or x is a negative rational number.

Proof. shcu

□

Problem 4 - A

For the properties of the order on the rationals, let $x, y, z \in \mathbb{Q}$. Then, the following properties hold.

Problem 5 - H

Show that if x, y, z are rational numbers such that $x < y$ and z is negative, then $xz > yz$.

Proof. chucu

□

C: Absolute Value and Exponentials

Problem 1 - Properties of absolute value and distance

Verify the basic properties of absolute value and distance. Let $x, y, z \in \mathbb{Q}$.

Proposition 1.1. *Non-degeneracy of absolute value.* We have $|x| \geq 0$. Also, $|x| = 0$ if and only if $x = 0$.

Proof. d

Proof. s

□

Proposition 1.2. *(Triangle inequality for absolute value)* We have $|x + y| \leq |x| + |y|$.

Proof. s

Proposition 1.5. *(Non-degeneracy of distance)* We have $d(x, y) \geq 0$. Also, $d(x, y) = 0$ if and only if $x = y$.

Proof. smth

□

Proposition 1.3. We have the inequalities $-y \leq x \leq y$ if and only if $y \geq |x|$. In particular, we have $-|x| \leq x \leq |x|$.

Proof. s

Proposition 1.6. *(Symmetry of distance)* We have $d(x, y) = d(y, x)$.

Proof. sh

□

Proposition 1.4. *(Multiplicativity of absolute value)* We have $|xy| = |x||y|$. In particular, $|-x| = |x|$.

Proposition 1.7. *(Triangle inequality for distance)* We have $d(x, z) \leq d(x, y) + d(y, z)$.

Proof. chu

□

Problem 2 - Properties of ε -closeness

Let $x, y, z, w \in \mathbb{Q}$. Then, the following properties hold:

Proposition 2.1. The number x is ε -close to y for every $\varepsilon > 0$ if and only if $x = y$.

Proof. s

Proof. chuchu

□

Proposition 2.2. Let $\varepsilon > 0$. If x is ε -close to y , then y is ε -close to x .

Proof. s

Proposition 2.5. Let $\varepsilon > 0$. If y and z are both ε -close to x and w is between y and z (i.e., $y \leq w \leq z$ or $z \leq w \leq y$), then w is also ε -close to x .

Proof. chuchu

□

Proposition 2.3. Let $\varepsilon, \delta > 0$. If x is ε -close to y , and y is δ -close to z , then x and z are $(\varepsilon + \delta)$ -close.

Proof. chuchu

Proposition 2.6. Let $\varepsilon > 0$. If x and y are ε -close, and z is non-zero, then xz and yz are $\varepsilon|z|$ -close.

Proof.

□

Proposition 2.4. Let $\varepsilon > 0$. If x and y are ε -close, they are also ε' -close for every $\varepsilon' > \varepsilon$.

Proposition 2.7. Let $\varepsilon, \delta > 0$. Suppose x and y are ε -close, and z and w are δ -close. Then xz and yw are $(\varepsilon|z| + \delta|w| + \varepsilon\delta)$ -close.

Problem 3 - Properties of Exponentiation I

Let x, y be rational numbers. Let n, m be natural numbers. The following properties hold.

Proposition 3.1. We have $x^n x^m = x^{n+m}$.

Proof. chuchu

Proof. suppose

□

Proposition 3.2. We have $(x^n)^m = x^{nm}$.

Proof. chuchu

Proposition 3.5. If $x \geq y \geq 0$, then $x^n \geq y^n \geq 0$.

Proof. suppose

□

Proposition 3.3. $(xy)^n = x^n y^n$.

Proof. chuchu

Proposition 3.6. If $x > y \geq 0$ and $n > 0$, then $x^n > y^n \geq 0$.

Proof. suppose

□

Proposition 3.4. Suppose $n \geq 0$. Then, we have $x^n = 0$ if and only if $x = 0$.

Proposition 3.7. We have $|x^n| = |x|^n$.

Proof. suppose

□

Problem 4 - Properties of Exponentiation II

Let x, y be rational numbers. Let n, m be integers. The following properties hold.

Proposition 4.1. We have $x^n x^m = x^{n+m}$.

Proof. suppose

□

Proof. chuchu

□

Proposition 4.2. We have $(x^n)^m = x^{nm}$.

Proposition 4.5. If $x, y > 0$, $n \neq 0$, and $x^n = y^n$, then $x = y$.

Proof. chuchu

□

Proof. suppose

□

Proposition 4.3. $(xy)^n = x^n y^n$.

Proof. chuchu

□

Proposition 4.6. We have $|x^n| = |x|^n$.

Proposition 4.4. If $x \geq y > 0$, then $x^n \geq y^n > 0$ if n is positive OR $0 < x^n \leq y^n$ if n is negative.

Proof. suppose

□

Problem 5 - Exponentiation grows faster than linear

Prove that $2^N \geq N$ for all positive integers N .

Proof. chuchu

□